

Sebastian Ashcallay Silva

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Data-focused software developer specializing in **data pipelines, machine learning, and analytics-driven systems**. Experienced in **cleaning, transforming, and analyzing large-scale datasets** using Python and SQL, with a focus on building **predictive models and data products**. Strong background in **agentic AI workflows, API integration, and data engineering**, with a passion for turning complex data into actionable insights for real-world applications.

EDUCATION

University of Maryland	College Park, MD
<i>Bachelor of Science in Computer Science (Machine Learning Track), Minor in Mathematics</i>	<i>Jan. 2024 – Dec. 2025</i>

PROFESSIONAL EXPERIENCE

Software Developer Intern	Jan. 2025 – Present
<i>Apollonian — Mental Wellness Platform</i>	<i>Remote</i>
– Led development of a multimodal video generation pipeline integrating LLM-based script generation, image synthesis, text-to-speech, and automated rendering, supporting 350+ daily users. – Built a CRM/outreach website integrating structured contact data, Agentic AI for assistance tools, and REST APIs, enabling automated data-driven messaging workflows and improving scalability of outreach operations. – Built an end-to-end data pipeline for outreach automation, ingesting and merging data from SAMHSA and NPPES APIs, performing web scraping and email enrichment, validating and scoring contact data, and storing structured results in Firestore for downstream filtering and campaign execution. – Developed a data filtering and experimentation system to curate high-quality outreach targets, applying rule-based validation and scoring logic, and enabling A/B testing workflows to evaluate engagement performance across messaging strategies. – Implemented Firebase Authentication and Firestore for secure user management and scalable NoSQL data storage; deployed frontend using Firebase Hosting. – Established CI/CD pipelines for streamlined deployment and iteration of company's CRM website. – Designed data schemas and storage pipelines using Firestore and Cloud Storage to manage structured and unstructured data, ensuring efficient retrieval for analytics and downstream processing. – Analyzed user interaction and content performance data to identify engagement patterns, supporting product decisions and improving user experience through data-driven insights. – Collaborated with leadership on product strategy and user acquisition, contributing to product demos and onboarding sessions, helping gain \$1.5k in investment from startup competitions (Pitch Dingman, XFoundry).	
Tutor	Sep. 2024 – Jul. 2025
<i>C2 Education</i>	<i>Bethesda, MD</i>
– Led 15+ weekly sessions across Math, Physics, and Computer Science, improving student performance in standardized testing. – Increased SAT Math scores from 580–620 to 700+ through structured learning plans and performance tracking.	

PROJECTS

Professor Ratings Predictor <i>Python, Pandas, Scikit-Learn, NLTK</i>	Feb. 2025 – Apr. 2025
– Developed a machine learning workflow analyzing 37,000+ student reviews across 4,100 professors, transforming unstructured text into structured predictive features. – Performed sentiment analysis using NLTK, engineering statistical features correlated with instructor ratings. – Trained and evaluated regression models (e.g., Ridge, SVR) with 10-fold cross-validation, achieving RMSE 0.452 and R^2 0.785. – Conducted feature importance analysis and dimensionality reduction (PCA) to interpret model behavior and generate insights from student feedback data.	
City Classification via CNNs <i>Python, PyTorch</i>	Mar. 2025 – May 2025
– Developed a CNN classification system to predict U.S. city locations from stereographic projections using transfer learning with ResNet18, achieving 95.25% precision and 100% Top-3 accuracy. – Designed preprocessing and augmentation pipelines, creating datasets with 2,700+ images to improve model robustness. – Implemented full training and evaluation workflow using PyTorch and Scikit-Learn, including normalization, F1 scores, and confusion matrix analysis. – Co-authored a detailed report summarizing methodology, experiments, and model performance insights.	

RESEARCH / WRITING

- Is Generative A.I. the Future of Education?** | EdTech & AI in K-12
- Feb. 2026
- Authored an analytical paper in English and Spanish examining generative AI in classrooms, focusing on learning retention, student engagement, and academic integrity.
 - Evaluated risks and proposed human-AI collaboration strategies to mitigate hallucinations and superficial learning.

CERTIFICATIONS

- AWS Cloud Practitioner**
- 2026
- Amazon Web Services*

TECHNICAL SKILLS

- Programming Languages:** Python, JavaScript, SQL, Java, C/C++, HTML, CSS
- AI / Machine Learning:** Machine Learning, Deep Learning, Generative AI, LLMs, Agentic AI, NLP, CNNs, Transfer Learning, Prompt Engineering, Model Evaluation, Feature Engineering, Sentiment Analysis
- Frameworks & Libraries:** FastAPI, React, Node.js, Scikit-Learn, PyTorch, NLTK
- Cloud & Backend Systems:** Google Cloud Platform (GCP), AWS, Firebase (Authentication, Firestore, Hosting, Functions, Cloud Storage), REST APIs, Serverless Architectures, Microservices
- Data Engineering & Analytics:** Pandas, NumPy, SQL, Data Cleaning, Data Transformation, ETL Pipelines, Data Validation, Exploratory Data Analysis, Feature Engineering, Web Scraping, API Integration
- DevOps & Workflow:** Git, CI/CD (GitHub Actions), Docker (basic), Linux/Bash, Asynchronous Processing, API Orchestration